

Stephen Murphy

I help businesses deliver products to market through solid engineering design, ambitious goal-setting, effective communication between project teams, and positive work cultures.

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📍 San Diego, CA, USA

WORK EXPERIENCE

Sr. Staff Mechanical Engineer - Mechanical Systems Shield AI

05/2025 - Present
San Diego, CA

- Owned the conceptual development of the X-BAT wing fold mechanism which supports full 180° folding of both wings to achieve an unprecedented spotting factor of 0.33 on a 39' wingspan aircraft inclusive of mechanism, actuators, plumbing, lock pins, and doors all within mass budget.
- Designed the environmental cooling system (ECS) for the X-BAT Hopper prototype including heat exchangers, plumbing, manifolds, fluid storage, instrumentation, and air cooling.
- Personally developed CAD and drawings in Siemens NX and simulations in Simcenter.

Staff Mechanical Engineer Shield AI

12/2023 - 04/2025
San Diego, CA

- Led the development of the V-BAT wing and fuselage modules from concept to production with novel designs for a cam-lock attachment mechanism, rotating pitot-static probe for easier storage, rib-free composite structure for reduced cost and assembly time, radio and antenna integrations, electrical harnessing, and PCBA integrations.

Mechanical Design Manager SkySafe

01/2023 - 11/2023
San Diego, CA

- Led the redesign of two drone detection platforms from concept to production for improved ease of manufacture, cost, and reliability. This included thermal and structural FEA, CFD, CAD, drawings per ASME Y14.5-2018, RoHS & CE compliance reviews, prototyping, and testing.

Senior Systems Engineer Reverb Industrial

09/2022 - 01/2023
San Diego, CA

- Defined system architecture and completed the full mechanical and electrical design of the first Alpha prototype for a novel wire-based metal additive-manufacturing platform.

Design Manager, Advanced & New Product Development SharkNinja

03/2021 - 09/2022
Boston, MA

- Owned the product design engineering of Shark FlexStyle, HyperAir, SpeedStyle, and SmoothStyle high volume consumer electronics products from PoC to mass production at volumes > 100k units/year
- Led and performed mechanical design work including CAD (CREO), electronics integration, injection molding, DFM, CFD, FEA, DFMEA, and tolerance analysis
- Promoted to US Personal Care Engineering Lead, where I led a team of 4 AD/NPD engineers and cross-functional representation

Software Engineering Manager / Software Engineer Intel

09/2018 - 01/2020
St. Leon-Rot, Germany

- Led a 9-person software team to the successful release of Intel Mission Control, a drone flight planning application for Windows, and personally programmed UIs in Java

Senior Mechanical Engineer Intel

10/2016 - 09/2018
St. Leon-Rot, Germany

- Developed a commercial UAS from scratch including a 2kg aircraft, batteries, GNSS base station, remote, and charger, resulting in my promotion to lead a team of 9 mechanical engineers

Mechanical Engineer MAVinci

10/2015 - 10/2016
St. Leon-Rot, Germany

- Designed and built a custom 3-axis CNC and a 2kg fixed wing UAS

STRENGTHS

Industries

- Aerospace
- Batteries
- Consumer electronics

CAD

- NX
- Creo
- SOLIDWORKS
- Fusion 360

Programming & Simulation

- MATLAB
- Python
- Ansys
- Java
- Simcenter 3D
- COMSOL

Mechanical Design

- Tolerance analysis
- Fluid systems
- Composites
- DFM/DFA
- Acoustics
- Vibration
- Batteries
- Actuators
- Injection molding
- DFMEA
- Motors
- Thermals

Leadership

- Team building
- Architecture
- Technical ownership
- Engineering management
- Cross-functional execution

EDUCATION

Master of Engineering in Mechanical Engineering

University of Victoria 2020

Project: Thermo-electrochemical Model of a Zinc-air Flow Battery

Bachelor of Engineering in Mechanical Engineering with Co-op

University of Victoria 2015

Top 15% of class; honors thesis on high-altitude, long-endurance biplane design

LANGUAGES

English: Native speaker

German: Limited working proficiency

PERSONAL INTERESTS

- Cooking while listening to old soul
- Building and riding road bikes
- Ultralight hiking
- Tennis, Swimming, Yoga, Running